



L1A: Introduction; Elements of Programming

CS1101S: Programming Methodology

Boyd Anderson

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What is CS1101S?

Course management

Elements of programming



What is CS1101S?

Course history

Course management

Elements of programming



The goals of this course

- ▶ Understand
the *structure and interpretation of computer programs*



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- ▶ Understand
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- ▶ Learn how to program



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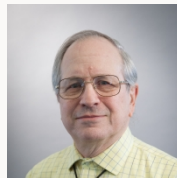
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Course origins

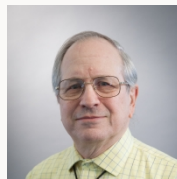
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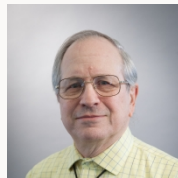


- ▶ Book: “Structure and Interpretation of Computer Programs”



Course origins

Course concept developed at MIT in the 1980s,
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- ▶ Book: “Structure and Interpretation of Computer Programs”
- ▶ A.K.A. the “SICP”



CS1101S Legacy: The Beginning

Introduced in DISCS in Sem 1 1997/1998 as “IC1101S”
Lecturers: Jacob Katzenelson and Leong Tze Yun



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Jacob renamed it to CS1101S in 1998/1999 Sem 1



CS1101S Today

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- ▶ Adapted SICP for JavaScript instead of Scheme
- ▶ Low Kok Lim joined in 2014. He teaches the semester two version of the course!
- ▶ Boyd joined in 2020 and Sanka joined in 2022.
- ▶ In 2026, we moved from JavaScript to Python.



What is CS1101S?

Course management

- Weekly flow

- People

- Course overview

- Resources

- Other Matters

Elements of programming



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The weekly flow



The weekly flow

- ▶ Wednesday 2-h **Lecture XA**: introduces new concepts, includes a **Check In**



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- ▶ Wednesday 2-h **L**ecture XA: introduces new concepts, includes a **C**heck In
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- ▶ Thursday 1-h **Reflection**: reflects on lecture material
- ▶ Friday 2-h **Lecture XB**: introduces new concepts, includes a **Check In**



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- ▶ Wednesday 2-h **L**ecture XA: introduces new concepts, includes a **C**heck In
- ▶ Wednesday **P**ath: online post-lecture quiz in Source Academy
- ▶ Thursday 1-h **R**eflection: reflects on lecture material
- ▶ Friday 2-h **L**ecture XB: introduces new concepts, includes a **C**heck In
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- ▶ Monday/Tuesday 2-h **S**tudio session: **Master the concepts!**



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- ▶ Monday/Tuesday 2-h **S**tudio session: **Master the concepts!**
- ▶ **M**issions, **Q**uests, **C**ontests: Show your mastery!



Lecture venues

We have two venues for lectures

- ▶ LT16
- ▶ LT38



Lecture venues

We have two venues for lectures

- ▶ LT16
- ▶ LT38

Please note

Cadets, report to **your** allocated venue and only your allocated venue.



Lecture venues

We have two venues for lectures

- ▶ LT16
- ▶ LT38

Please note

Cadets, report to **your** allocated venue and only your allocated venue.

The teaching staff will alternate between the two venues to ensure that all cadets get a live lecture experience.



Homework to be submitted throughout the semester

Throughout the semester, we will have...



Homework to be submitted throughout the semester

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- ▶ 17 Paths



Homework to be submitted throughout the semester

Throughout the semester, we will have...

- ▶ 17 Paths
- ▶ 20 Missions



Homework to be submitted throughout the semester

Throughout the semester, we will have...

- ▶ 17 Paths
- ▶ 20 Missions
- ▶ 15 Quests



Homework to be submitted throughout the semester

Throughout the semester, we will have...

- ▶ 17 Paths
- ▶ 20 Missions
- ▶ 15 Quests

(some adjustment possible)



This week



This week

- ▶ today 2-h **L**ecture L1A: Elements of Programming



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- ▶ today 2-h **L**ecture L1A: Elements of Programming
- ▶ today **P**ath P1A on Elements of Programming



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- ▶ today 2-h **L**ecture L1A: Elements of Programming
- ▶ today **P**ath P1A on Elements of Programming
- ▶ tomorrow 1-h **R**eflection R1: choose between 10am and 2pm sessions



This week

- ▶ today 2-h **L**ecture L1A: Elements of Programming
- ▶ today **P**ath P1A on Elements of Programming
- ▶ tomorrow 1-h **R**eflection R1: choose between 10am and 2pm sessions
- ▶ Friday 2-h **L**ecture L1B: Runology



This week

- ▶ today 2-h **L**ecture L1A: Elements of Programming
- ▶ today **P**ath P1A on Elements of Programming
- ▶ tomorrow 1-h **R**eflection R1: choose between 10am and 2pm sessions
- ▶ Friday 2-h **L**ecture L1B: Runology
- ▶ Friday **P**ath P1B on Runology



This week

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- ▶ today **P**ath P1A on Elements of Programming
- ▶ tomorrow 1-h **R**eflection R1: choose between 10am and 2pm sessions
- ▶ Friday 2-h **L**ecture L1B: Runology
- ▶ Friday **P**ath P1B on Runology
- ▶ Friday Mission “Rune Trials” comes out, due on Tuesday, 18/8



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- ▶ today 2-h **L**ecture L1A: Elements of Programming
- ▶ today **P**ath P1A on Elements of Programming
- ▶ tomorrow 1-h **R**eflection R1: choose between 10am and 2pm sessions
- ▶ Friday 2-h **L**ecture L1B: Runology
- ▶ Friday **P**ath P1B on Runology
- ▶ Friday Mission “Rune Trials” comes out, due on Tuesday, 18/8
- ▶ Monday/Tuesday 2-h **S**tudio session S2 on Elements of Programming



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Lecturers



Lecturers

Their roles

Conduct **lectures** on Wednesdays and on Fridays, coordinate the course



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Conduct **lectures** on Wednesdays and on Fridays, coordinate the course



Martin Henz



Sanka Rasnayaka



Boyd Anderson



Tutors

Their role

Facilitate weekly **Reflection** sessions



Tutors

Their role

Facilitate weekly **Reflection** sessions

Eldric Liew Chun Tze

Enzio Kam Hai Hong

Zhang Yao

Martin Henz, Sanka Rasnayaka, Boyd Anderson



Source Academy Team

Their roles

Design, develop, deploy, maintain the **Source Academy**, our learning environment for CS1101S

Richard Dominick, Source Academy CTO

Zhang Yao, Frontend.



Staff Photos

The team





Avengers

Their role

- ▶ facilitating weekly 2-h Studio sessions,
- ▶ guiding you in the fine art of programming,
- ▶ discussing your missions and quests with you and grading them
- ▶ conducting mastery checks (more on this later)



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Course Overview

- ▶ Unit 1—Functions (textbook Chapter 1)
 - Getting acquainted with the elements of programming, using **functional abstraction**
 - Learning to read programs, and using the **substitution model**
 - Example applications: Runes, curves
- ▶ Unit 2—Data (textbook Chapter 2)
 - Getting familiar with data: pairs, lists, trees, streams
 - Searching in lists and trees, sorting of lists
 - Example application: sound processing



Course Overview, continued

- ▶ Unit 3—State (parts of textbook Chapter 3)
 - Programming with stateful abstractions
 - Arrays, loops, searching in and sorting of arrays
 - Reading programs using the environment model / CSE machine
 - Example applications: robotics, video processing
- ▶ Unit 4—Beyond (parts of textbook Chapters 3 and 4)
 - Memoization, Memoized Streams
 - Metalinguistic abstraction
 - Objects and Types



Special Event: Sumobot



Wednesday, October 14th, 2026. Details to come.



Assessment timeline



Assessment timeline

- ▶ Reading **A**ssessment RA1 on Friday Week 4: 4/9 (Unit 1)



Assessment timeline

- ▶ **Reading Assessment RA1** on Friday Week 4: 4/9 (Unit 1)
- ▶ **Programming Assessment PA1** on Friday Week 6: 18/9 (Unit 1)



Assessment timeline

- ▶ **Reading Assessment RA1** on Friday Week 4: 4/9 (Unit 1)
- ▶ **Programming Assessment PA1** on Friday Week 6: 18/9 (Unit 1)
- ▶ **Recess Week**



Assessment timeline

- ▶ **Reading Assessment** RA1 on Friday Week 4: 4/9 (Unit 1)
- ▶ **Programming Assessment** PA1 on Friday Week 6: 18/9 (Unit 1)
- ▶ **Recess Week**
- ▶ **Midterm** Assessment on Wednesday Week 7: 30/9 (Unit 2)



Assessment timeline

- ▶ **Reading Assessment RA1** on Friday Week 4: 4/9 (Unit 1)
- ▶ **Programming Assessment PA1** on Friday Week 6: 18/9 (Unit 1)
- ▶ **Recess Week**
- ▶ **Midterm** Assessment on Wednesday Week 7: 30/9 (Unit 2)
- ▶ **Reading Assessment RA2** on Friday Week 10: 23/10 (Unit 3)



Assessment timeline

- ▶ **Reading Assessment RA1** on Friday Week 4: 4/9 (Unit 1)
- ▶ **Programming Assessment PA1** on Friday Week 6: 18/9 (Unit 1)
- ▶ **Recess Week**
- ▶ **Midterm Assessment** on Wednesday Week 7: 30/9 (Unit 2)
- ▶ **Reading Assessment RA2** on Friday Week 10: 23/10 (Unit 3)
- ▶ **Programming Assessment PA2** on Friday Week 12: 6/11 (Unit 4)



Assessment timeline

- ▶ **Reading Assessment RA1** on Friday Week 4: 4/9 (Unit 1)
- ▶ **Programming Assessment PA1** on Friday Week 6: 18/9 (Unit 1)
- ▶ **Recess Week**
- ▶ **Midterm Assessment** on Wednesday Week 7: 30/9 (Unit 2)
- ▶ **Reading Assessment RA2** on Friday Week 10: 23/10 (Unit 3)
- ▶ **Programming Assessment PA2** on Friday Week 12: 6/11 (Unit 4)
- ▶ **Final Assessment:** 25/11



Assessment components (1/2)

5%: Source Academy XP (Missions, Quests, Paths, Studios, ...): 24,000 XP



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- 3%: Check In



Assessment components (2/2)

8%: Reading Assessment 1, Week 4



Assessment components (2/2)

8%: Reading Assessment 1, Week 4

8%: Programming Assessment 1, Week 6



Assessment components (2/2)

8%: Reading Assessment 1, Week 4

8%: Programming Assessment 1, Week 6

4%: Mastery Check 1, any time



Assessment components (2/2)

- 8%: Reading Assessment 1, Week 4
- 8%: Programming Assessment 1, Week 6
- 4%: Mastery Check 1, any time
- 20%: Midterm Assessment, Week 7



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- 20%: Midterm Assessment, Week 7
- 8%: Reading Assessment 2, Week 10



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 - 30%: Final Assessment
- (Wait! That's 108%...)*



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 - 20%: Midterm Assessment, Week 7
 - 8%: Reading Assessment 2, Week 10
 - 8%: Programming Assessment 2, Week 12
 - 4%: Mastery Check 2, any time
 - 30%: Final Assessment
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- Best 3 of 4 for RA/PA!*



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Your homework

Homework (Paths, Missions, Quests, Studios) earn XP.



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Contests

Look out for announcements. Winners & runners-up receive XP.



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Miscellaneous special topics

Extra XP to deserving individuals; look out for announcements.



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Resources

Textbook: SicPy

Text Book can be found here: [SicPy](#)



Resources

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Course Schedule and Important Links

Canvas > [CS1101S](#)



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Course FAQs can be found here

Canvas > CS1101S > [Frequently Asked Questions](#)



Need help?



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Many of you are signed up, already. If you cannot sign up, please email Martin



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- ▶ PM your Avenger (WhatsApp/FB/Telegram/you-name-it:
check what channel they prefer)



Need help?



Need help?

- ▶ Email your Reflection instructor:
Canvas > CS1101S > [Teaching Staff](#)



Need help?

- ▶ Email your Reflection instructor:
Canvas > CS1101S > Teaching Staff
- ▶ Email lecturers
 - Martin: henz@comp.nus.edu.sg
 - Sanka: sanka@nus.edu.sg
 - Boyd: boyd@comp.nus.edu.sg



Academic Honesty

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Be sure read the [Code of Student Conduct](#)



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- ▶ These AI tools will be a great resource for you in your learning journey. Make sure you try them out. Ask your Avenger how they are using them.
- ▶ Good news: These AI tools know our material very well. There might still be mistakes in their answers, so make sure you verify the answers by asking your Avenger, tutor, or lecturer.



Our AI Policy

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- ▶ You can however use them for preparation for your reflection session. Coming up tomorrow!



What is CS1101S?

Course management

Elements of programming

- Processes and programming

- Primitive expressions, 1.1.1

- Operator combinations, 1.1.1

- Evaluating combinations, 1.1.3

- Naming abstraction, 1.1.2

- Functional abstraction, 1.1.4

- Predicates and conditional expressions, 1.1.6



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Processes and programming

Primitive expressions, 1.1.1

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Naming abstraction, 1.1.2

Functional abstraction, 1.1.4

Predicates and conditional expressions, 1.1.6



Processes



Processes

Definition (from Wikipedia)

A process is a set of activities that interact to produce a result. The activities unfolds according to patterns that *describe* or *prescribe* the process.



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A process is a set of activities that interact to produce a result. The activities unfolds according to patterns that *describe* or *prescribe* the process.

Examples

Processes are everywhere. They permeate our nature and culture:

- ▶ Galaxies and solar systems follow *formation* processes.
- ▶ Metabolic pathways in our bodies are *biological* processes.
- ▶ Political parties, legislature, courts, etc. follow processes.
- ▶ Industrial production is governed by processes.



Computational processes

Definition

A *computational process* is a set of activities in a computer, designed to achieve a desired result.



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Our task

Here we are concerned about how this *design* happens.



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Our task

Here we are concerned about how this *design* happens.

Our design method

We use *programs* to prescribe how the computational process unfolds.



What is programming?



What is programming?

People in focus

As the complexity of computer systems increases, *communication* between architects, designers, programmers, operators and users becomes increasingly important.



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As the complexity of computer systems increases, *communication* between architects, designers, programmers, operators and users becomes increasingly important.

Programs as communication devices

Since programs prescribe the computational processes at the heart of these systems, they can play a central role in the *human* communication during their construction and operation.



What is programming?

People in focus

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Since programs prescribe the computational processes at the heart of these systems, they can play a central role in the *human* communication during their construction and operation.

Our motto

Programming is communicating computational processes.



Our programming environment: Source Academy



Our programming environment: Source Academy

Website designed for CS1101S



Our programming environment: Source Academy

Website designed for CS1101S

Has just what you need for understanding the *structure and interpretation of computer programs*.



Our programming environment: Source Academy

Website designed for CS1101S

Has just what you need for understanding the *structure and interpretation of computer programs*.

Research

Source Academy is also an *educational research tool*: We want to study what happens when people like you learn how to program.



Your first login to Source Academy



Your first login to Source Academy

Consent

You can be part of our research by letting us use your programs anonymously.



Your first login to Source Academy

Consent

You can be part of our research by letting us use your programs anonymously. But if you don't want to: No worries: There will be **no penalty** if you don't consent.



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Our ship

You think you are studying at NUS?



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You think you are studying at NUS? Wrong! You are cadets in...



Your first login to Source Academy

Consent

You can be part of our research by letting us use your programs anonymously. But if you don't want to: No worries: There will be **no penalty** if you don't consent.

Our ship

You think you are studying at NUS? Wrong! You are cadets in... the Source Academy!



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Visit <https://sourceacademy.nus.edu.sg>, decide on consent and click on "Playground". (Source Academy opens later in the lecture.)



What is CS1101S?

Course management

Elements of programming

Processes and programming

Primitive expressions, 1.1.1

Operator combinations, 1.1.1

Evaluating combinations, 1.1.3

Naming abstraction, 1.1.2

Functional abstraction, 1.1.4

Predicates and conditional expressions, 1.1.6



Elements of programming



Elements of programming

Every powerful language provides...



Elements of programming

Every powerful language provides...

- ▶ Primitives



Elements of programming

Every powerful language provides...

- ▶ Primitives
- ▶ Combination



Elements of programming

Every powerful language provides...

- ▶ Primitives
- ▶ Combination
- ▶ Means of abstraction



Primitive expressions



Primitive expressions

- ▶ Numerals: 0, -42, 486



Primitive expressions

- ▶ Numerals: 0, -42, 486
- ▶ Numerals use decimal notation



Primitive expressions

- ▶ Numerals: 0, -42, 486
- ▶ Numerals use decimal notation
- ▶ Our interpreter can evaluate numerals, resulting in the numbers they represent



A detail

Evaluating an expression alone shows nothing

If you give the interpreter just 486, it responds by printing — nothing.



A detail

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If you give the interpreter just 486, it responds by printing — nothing.

Example

To see the result, apply the primitive function `print` to it:
`print(486)`



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Means of Combination: Operators

Examples

```
print(5 * 99)
```

```
print(25 - (4 + 2) * 3)
```





Means of Combination: Operators

Examples

```
print(5 * 99)
```

```
print(25 - (4 + 2) * 3)
```



Notation as usual

operator between operands: *infix notation* with *precedences*



Evaluating Operator Combinations

But exactly how...

...do we evaluate

```
print((2 + 4 * 6) * (3 + 12))
```





Evaluation of expressions

Demonstration: Evaluation of expressions



Evaluation of expressions

1 + 2 * 3 + 4





Evaluation of expressions

1 + 2 * 3 + 4



Replace “eligible” subexpression with result

(1 + (2 * 3)) + 4



Evaluation of expressions

$1 + 2 * 3 + 4$



Replace “eligible” subexpression with result

$(1 + (2 * 3)) + 4$

$(1 + 6) + 4$



Evaluation of expressions

1 + 2 * 3 + 4



Replace “eligible” subexpression with result

(1 + (2 * 3)) + 4

(1 + 6) + 4

7 + 4



Evaluation of expressions

$1 + 2 * 3 + 4$



Replace “eligible” subexpression with result

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Means of Abstraction: Naming

Example

```
size = 2  
print(5 * size)
```





Evaluation of declaration assignments

```
a = 3 * 5  
print(a + 5)
```





Evaluation of declaration assignments

```
a = 3 * 5  
print(a + 5)
```



Evaluate what is after the =

```
size = 3  
2 * size
```



Evaluation of declaration assignments

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Then replace that name in rest of the program with that value.

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Evaluation of declaration assignments

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Evaluate what is after the =

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```

```
6
```



Primitive Names

Primitive constants

Python has a few primitive names. For example, the name `math.pi` refers to the constant π .



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Primitive functions

In addition to operators such as `+`, Python has primitive functions. For example, `math.sqrt` is the square root function.



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Primitive functions

In addition to operators such as `+`, Python has primitive functions. For example, `math.sqrt` is the square root function.

Function application expressions

Functions can be applied as usual in mathematics, for example

```
print(math.sqrt(15))
```



applies the primitive square root function to 15.



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Means of Abstraction: Compound functions

Example

```
def square(x):  
    return x * x
```





Means of Abstraction: Compound functions

Example

```
def square(x):  
    return x * x
```



What does this statement mean?

Like declaration assignments, this *function definition* declares a name, here `square`. The value associated with the name is a function that takes an argument `x` and produces (returns) the result of multiplying `x` by itself.



Formatting matters: Indentation

Python uses indentation, not braces

Unlike some languages which use `{}`, Python uses *indentation* to show which statements belong to a function's body.



Formatting matters: Indentation

Python uses indentation, not braces

Unlike some languages which use {}, Python uses *indentation* to show which statements belong to a function's body.

Correctly indented

```
def square(x):  
    return x * x  
  
print(square(5))
```





Formatting matters: Indentation (continued)

Incorrectly indented

```
def square(x):  
return x * x
```



Formatting matters: Indentation (continued)

Incorrectly indented

```
def square(x):  
return x * x
```

Python can't tell that `return x * x` belongs inside `square` — so it refuses to guess, and raises an error instead.



Review: Function application

Recall from the `math_sqrt` example

We apply a function by supplying its arguments in parentheses.



Review: Function application

Recall from the `math_sqrt` example

We apply a function by supplying its arguments in parentheses.

Example

```
def square(x):  
    return x * x  
  
print(square(14))
```





More examples

```
print(square(21))  
  
print(square(2 + 5))  
  
print(square(square(3)))
```





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Boolean values

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The two boolean values `True` and `False` represent *answers* to yes-no questions



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A *predicate* is a function or an expression that returns or evaluates to a boolean value.



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Predicates

A *predicate* is a function or an expression that returns or evaluates to a boolean value.

Examples

```
True  
False  
x >= 1
```



Another predicate

Example

```
def adult(x):  
    return x >= 18
```





Conditional expressions

Why?

We would like to check something, e.g. answer a yes/no question, and return a different value, depending on the answer



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Back to the example

```
enter_club() if adult(my_age) else  
    go_to_movie()
```



Conditional expressions

Why?

We would like to check something, e.g. answer a yes/no question, and return a different value, depending on the answer

Back to the example

```
enter_club() if adult(my_age) else  
    go_to_movie()
```

Example: abs function

To compute the absolute value of a number, check if it is greater or equal 0. If yes, return the number unchanged, and if no, return the number negated.



Time for the Check In!



Time for the Check In!

Still here? Time to head to the Source Academy and check in!



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Look for our first Check In in the Source Academy!



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Welcome on board!